THE MINIMUM AND MAXIMUM FORCING NUMBERS OF BIPARTITE GRAPHS: AN ANSWER TO A QUESTION OF LIU, DIWAN AND ZHANG

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Abstract. For a perfect matching M of a graph G, the forcing number $f(G, M)$ is the smallest size of a subset of M contained in no other perfect matching; the minimum and maximum of $f(G, M)$ over all perfect matchings are denoted $f(G)$ and $F(G)$. The notion goes back to the "innate degree of freedom" of Kekulé structures in chemistry. Liu, Diwan and Zhang asked whether every bipartite graph G of order $2n$ with $F(G) \geq n - k$, where $1 \leq k \leq n - 1$, satisfies $f(G) \geq \lceil n/k \rceil - 1$; they proved this for $k = 1, 2$. We answer the question affirmatively for all k. Equivalently, every bipartite graph of order $2n$ with a perfect matching satisfies

$$n \leq \big(f(G) + 1\big)\big(n - F(G)\big).$$

The proof combines the triangular structure of a bipartite graph with a unique perfect matching, an interval representation of an arbitrary perfect matching, and the greedy algorithm for interval scheduling. The result has been formally verified in Lean 4 with Mathlib.

1. Introduction

Let G be a finite simple graph with a perfect matching M. A subset $S \subseteq M$ is a *forcing set* of M if no other perfect matching of G contains S, and the *forcing number* $f(G, M)$ is the smallest size of a forcing set of M. The concept originates in chemistry: perfect matchings of the graph of a benzenoid hydrocarbon are its Kekulé structures, and Klein and Randić [2] introduced the *innate degree of freedom* of a Kekulé structure, the forcing number in today's terminology [1]. The *minimum* and *maximum forcing numbers* of G are

$$f(G) = \min_M f(G, M), \qquad F(G) = \max_M f(G, M),$$

the extrema being over all perfect matchings. If G has order $2n$ then $0 \leq f(G) \leq F(G) \leq n - 1$.

Liu, Diwan and Zhang [3] studied graphs whose maximum forcing number is large. They showed that for a bipartite graph G of order $2n$, $F(G) = n - 1$ holds only for $K_{n,n}$, and that $F(G) = n - 2$ implies $f(G) \geq \lceil n/2 \rceil - 1$, and they proposed the following problem [3, Problem 5.5].

Problem 1.1 (Liu–Diwan–Zhang)**.** *Let G be a bipartite graph of order $2n$ and with $F(G) \geq n - k$ for $1 \leq k \leq n - 1$. Does $f(G) \geq \lceil n/k \rceil - 1$ always hold?*

We answer Problem 1.1 affirmatively.

Theorem 1.2. *Let G be a finite bipartite graph of order $2n$ with a perfect matching. Then*

$$n \leq \big(f(G) + 1\big)\big(n - F(G)\big).$$

Corollary 1.3. *The answer to Problem 1.1 is yes: if $F(G) \geq n - k$ with $1 \leq k \leq n - 1$, then $f(G) \geq \lceil n/k \rceil - 1$.*

Date: October 29, 2026.

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Proof. We have $n - F(G) \le k$, so Theorem 1.2 gives $n \le k\,(f(G)+1)$, i.e. $f(G)+1 \ge \lceil n/k \rceil$. □

Conversely, Corollary 1.3 applied with $k = n - F(G)$ gives Theorem 1.2 whenever $F(G) \ge 1$, and for $F(G) = 0$ the theorem is trivial; so the two statements are equivalent, and Theorem 1.2 is the answer to Problem 1.1 for all k at once. Connectedness of G is not needed.

The bound is attained. By exhaustive computation over all bipartite graphs with $n \le 5$ and random sampling for $6 \le n \le 9$, equality $n = (f(G)+1)(n - F(G))$ occurs for non-complete graphs, for instance with $(n, f, F) = (4,1,2)$, $(6,1,3)$, $(6,2,4)$, $(8,1,4)$, $(8,3,6)$ and $(9,2,6)$; and the graphs constructed in [3, Remark 5.6] satisfy $F(G) = n - k$ and $f(G) = \lceil n/k \rceil - 1$ in all 14 cases with $n \le 10$ that we checked.

2. Two lemmas

Throughout, G is a finite simple graph and a perfect matching is a set of pairwise disjoint edges covering every vertex.

Lemma 2.1. *Let M be a perfect matching of G and $S \subseteq M$. Then S is a forcing set of M if and only if the graph $G - V(S)$ obtained by deleting the endpoints of S has a unique perfect matching.*

Proof. The perfect matchings of G containing S are exactly the sets $S \cup N$ with N a perfect matching of $G - V(S)$; one of them is M, obtained from $N = M \setminus S$. □

The following description of bipartite graphs with a unique perfect matching is classical (see [4, Ch. 4]); we include the short proof.

Lemma 2.2 (Triangular ordering)**.** *Let H be a finite bipartite graph with a unique perfect matching $\{a_i b_i : i \in I\}$, the vertices a_i and b_i lying on opposite sides. Then the index set can be numbered $1, \dots, m$ so that every edge $a_i b_j$ of H satisfies $i \le j$.*

Proof. Put an arc $i \to j$ on I whenever $i \ne j$ and $a_i b_j$ is an edge. If $i_1 \to i_2 \to \cdots \to i_t \to i_1$ were a directed cycle, replacing the edges $a_{i_s} b_{i_s}$ by $a_{i_s} b_{i_{s+1}}$ (indices mod t) would give a second perfect matching. Hence the digraph is acyclic, and a topological ordering is the required numbering. □

3. Proof of Theorem 1.2

Let $s = f(G)$, and choose a perfect matching M_0 with a forcing set S_0 of size s. Let (A, B) be a bipartition of G; a perfect matching is a bijection between A and B, so $|A| = |B| = n$. Deleting the endpoints of S_0 removes s vertices from each side, and by Lemma 2.1 the induced subgraph $H = G - V(S_0)$ has the unique perfect matching $M_0 \setminus S_0$. Let $m = n - s$ and use Lemma 2.2 to label the vertices of H as $a_1, \dots, a_m \in A$ and $b_1, \dots, b_m \in B$ so that every edge $a_i b_j$ of H, hence (since H is an induced subgraph) every edge of G between these vertices, satisfies $i \le j$. Define coordinates

$$L(a_i) = i, \quad R(b_i) = i \quad (1 \le i \le m),$$
$$L(a) = 0 \ \ (a \in A \cap V(S_0)), \qquad R(b) = m+1 \ \ (b \in B \cap V(S_0)).$$

Now let M be an arbitrary perfect matching of G. To each edge $e = ab \in M$ with $a \in A$, $b \in B$ assign the integer interval $I_e = [L(a), R(b)]$. It is nonempty: if both endpoints lie in H this is the triangularity, and otherwise $L(a) = 0$ or $R(b) = m+1$.

Every integer lies in at most $s+1$ of these intervals. Fix $1 \le t \le m$. The intervals with $L \le t$ are those of the M-edges at the s deleted vertices of A and at $a_1, \dots, a_t$, so there are $s + t$ of them. The intervals with $R < t$ are those of the M-edges at $b_1, \dots, b_{t-1}$; there are $t - 1$ of them, and each also has $L \le t$ because it is nonempty. Hence exactly $(s+t) - (t-1) = s+1$ intervals contain t. The integers 0 and $m+1$ lie in exactly s intervals each, and other integers in none.

Greedy selection. Repeat the following until no interval is left: choose a remaining interval I_e with the smallest right endpoint r, put e into T, and remove all remaining intervals that contain r (including I_e). Every remaining interval has right endpoint at least r, so the removed intervals are exactly those with left endpoint at most r, and the intervals left for later steps start after r. Consequently the intervals I_e, $e \in T$, are pairwise disjoint, and since each step removes at most $s+1$ intervals,

$$n = |M| \le (s+1)\,|T|. \tag{1}$$

$M \setminus T$ forces M. List the edges of T as $a'_1b'_1, \dots, a'_tb'_t$ with their intervals from left to right, so that $R(b'_i) < L(a'_j)$ for $i < j$. These strict inequalities imply that b'_i and a'_j lie in H (a deleted vertex of B has the maximal coordinate $m+1$, and a deleted vertex of A has coordinate 0), and then triangularity shows that $a'_jb'_i$ is not an edge of G for $i < j$. Let N be a perfect matching of G containing $M \setminus T$. The edges of $M \setminus T$ cover every vertex outside $V(T)$, so the remaining edges $N \setminus (M \setminus T)$ form a perfect matching between $\{a'_1, \dots, a'_t\}$ and $\{b'_1, \dots, b'_t\}$; write them as $a'_ib'_{\sigma(i)}$ for a permutation σ. Since there are no edges $a'_jb'_i$ with $i < j$, we have $\sigma(i) \ge i$ for all i, hence σ is the identity and $N = M$.

Therefore $f(G, M) \le |M \setminus T| = n - |T|$, and (1) gives $n \le (s+1)|T| \le (s+1)\,(n - f(G, M))$. Choosing M with $f(G, M) = F(G)$ proves the theorem. □

4. Formal verification

The statement of Problem 1.1 was formalized in Lean 4 (version 4.33.1) with Mathlib, following the definitions of [3]:

```
def IsPerfectMatching (G : SimpleGraph V) (M : Finset (Sym2 V)) : Prop :=
  (∀ e ∈ M, e ∈ G.edgeSet) ∧ ∀ v, ∃! w, s(v, w) ∈ M
def IsForcingSet (G : SimpleGraph V) (M S : Finset (Sym2 V)) : Prop :=
  S ⊆ M ∧ ∀ N, IsPerfectMatching G N → S ⊆ N → N = M
noncomputable def forcingNumber (G : SimpleGraph V) (M : Finset (Sym2 V)) : ℕ :=
  sInf {r | ∃ S, IsForcingSet G M S ∧ S.card = r}
-- minForcingNumber / maxForcingNumber: sInf / sSup over perfect matchings

theorem problem55 (G : SimpleGraph V) (n k : ℕ)
    (hG : G.IsBipartite) (horder : Fintype.card V = 2 * n)
    (hpm : ∃ M, IsPerfectMatching G M)
    (hk : 1 ≤ k) (hkn : k ≤ n - 1)
    (hF : n - k ≤ maxForcingNumber G) :
    (n + k - 1) / k - 1 ≤ minForcingNumber G
```

Here V is a finite type with decidable equality, and $\lceil n/k \rceil$ is written as `(n + k - 1) / k`. The theorem is proved without `sorry` or `native_decide` and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`; the statement was also checked with Comparator against a separately fixed statement file. The formalization is the file `lean/KOS/GraphTheory/ForcingNumbers/Main.lean` of the repository.

Author contributions. The problem was selected by a Claude Opus 5.5 agent searching the chemistry and mathematical biology literature for open problems. GPT-6-Astra found the proof and wrote the Lean formalization. Claude Fable 5.1 refereed the proof against the definitions of [3], re-checked the formalization, and carried out the exhaustive and random computations. Claude Opus 5.5 checked the proof and the formalization and wrote the manuscript. A human operator initiated the project and provided computing resources; the mathematics, the formal proofs and the text were produced by the models.

References

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